Height Growth Equations for CONUS: Douglas-fir

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Data

We took the FIA height growth data subset for Douglas-fir, filtered out stray (likely keypunch errors) states (NY, PA, IL, and MI), FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 151791 growth observations.

The observations are distributed among L3 Codes as follows:

NA_L3NAME	NA_L3CODE	n
Cascades	6.2.7	36435
Coast Range	7.1.8	25303
Klamath Mountains	6.2.11	17284
Columbia Mountains/Northern Rockies	6.2.3	16686
Middle Rockies	6.2.10	10755
Blue Mountains	6.2.9	8624
North Cascades	6.2.5	7824
Idaho Batholith	6.2.15	5196
Southern Rockies	6.2.14	5019
Eastern Cascades Slopes and Foothills	6.2.8	4012
Strait of Georgia/Puget Lowland	7.1.7	3080
Sierra Nevada	6.2.12	2879
Canadian Rockies	6.2.4	1972
Willamette Valley	7.1.9	1462
Wasatch and Uinta Mountains	6.2.13	1269
Arizona/New Mexico Mountains	13.1.1	1163
Northwestern Great Plains	9.3.3	647
Colorado Plateaus	10.1.6	527
Northern Basin and Range	10.1.3	492

NA_L3NAME	NA_L3CODE	n
California Coastal Sage, Chaparral, and Oak Woodlands	11.1.1	473
Columbia Plateau	10.1.2	368
NA	NA	89
Central Basin and Range	10.1.5	69
Madrean Archipelago	12.1.1	47
Northwestern Glaciated Plains	9.3.1	38
Wyoming Basin	10.1.4	36
Southwestern Tablelands	9.4.3	14
Atlantic Coastal Pine Barrens	8.5.4	12
Northern Allegheny Plateau	8.1.3	9
Ridge and Valley	8.4.1	3
Northern Appalachian and Atlantic Maritime Highlands	5.3.1	2
Northern Piedmont	8.3.1	1
Snake River Plain	10.1.8	1

Height Growth Prediction

We fit the equation:

$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl + \beta_5 PPT_{at} + \beta_6 PPT_{sm} + \beta_7 EMT)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \tag{1}$$

to the all 151791 growth observations, estimating β_0 through β_6. Where:

- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- PPT_{at} = autumn precipitation (mm)
- PPT_{sm} = summer precipitation (mm)
- EMT = extreme minimum temperature ($^\circ C$)

ht_{max} was estimated by finding the maximum height observed in the data set for Douglas-fir. In our data that value is: 285 feet.

The parameter estimates and fit statistics are:

	Coef.	Std. error	t-stat.	p
b1	0.0137317	0.0001092	125.73421	0
b2	1.9117041	0.0155443	122.98408	0
b3	0.2525859	0.0057994	43.55356	0
b4	0.0015453	0.0000319	48.43954	0

	Coef.	Std. error	t-stat.	p
b5	0.0008611	0.0000119	72.59500	0
b6	-0.0006749	0.0000638	-10.57578	0
b7	0.0249814	0.0002783	89.76713	0

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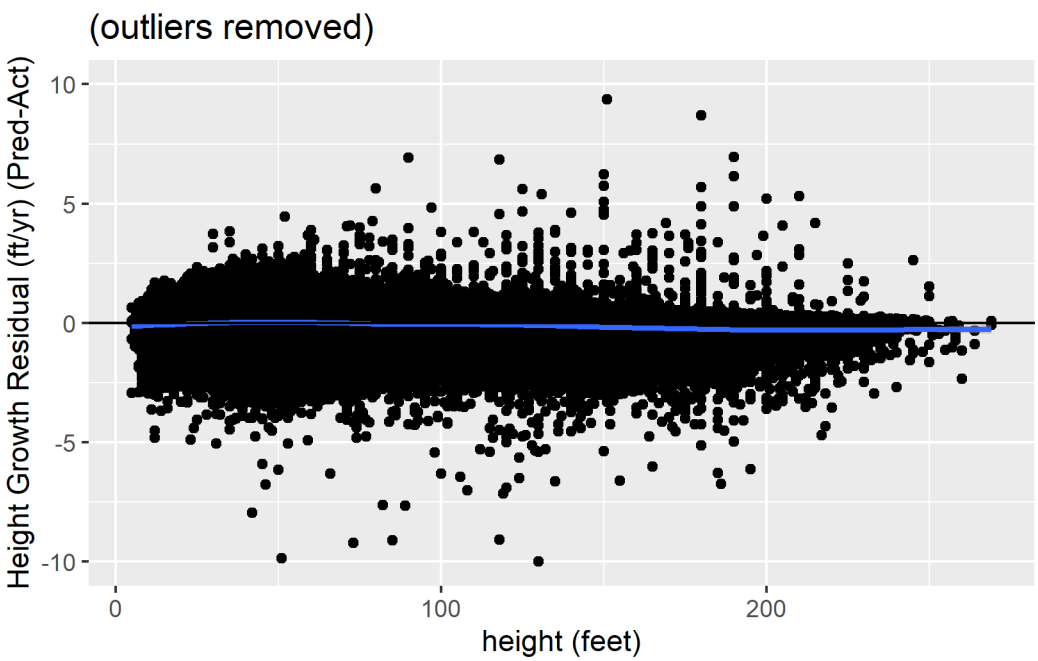
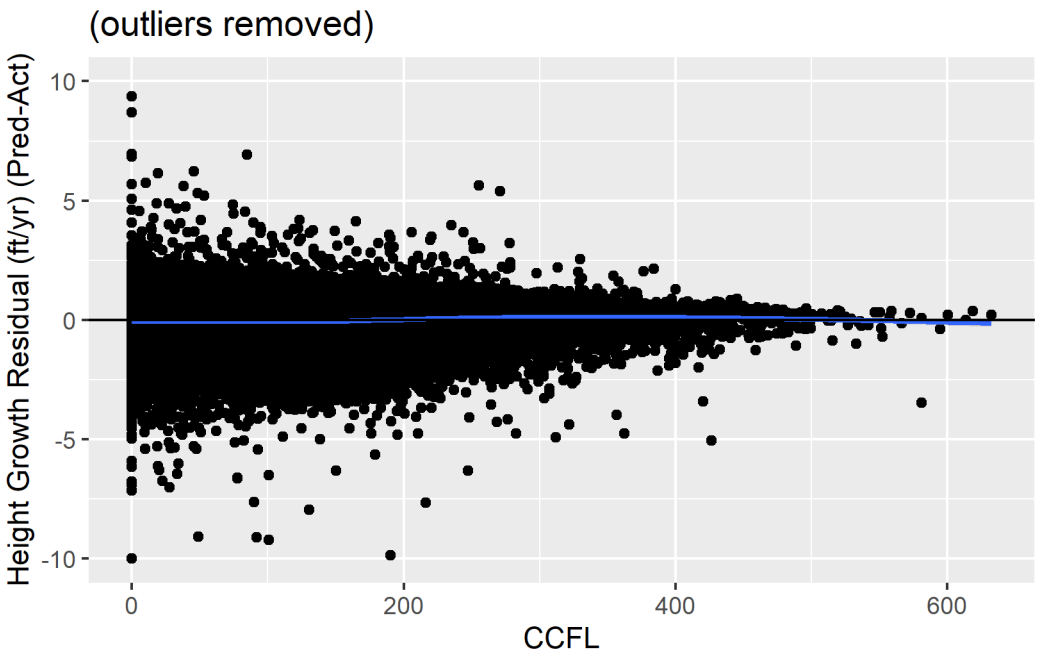
Residual Standard Error: 6.92751822680315 on 151784 degrees of freedom. AIC =  1018356.7
```

Residual Analysis for Equation 1

Mensurational Variables

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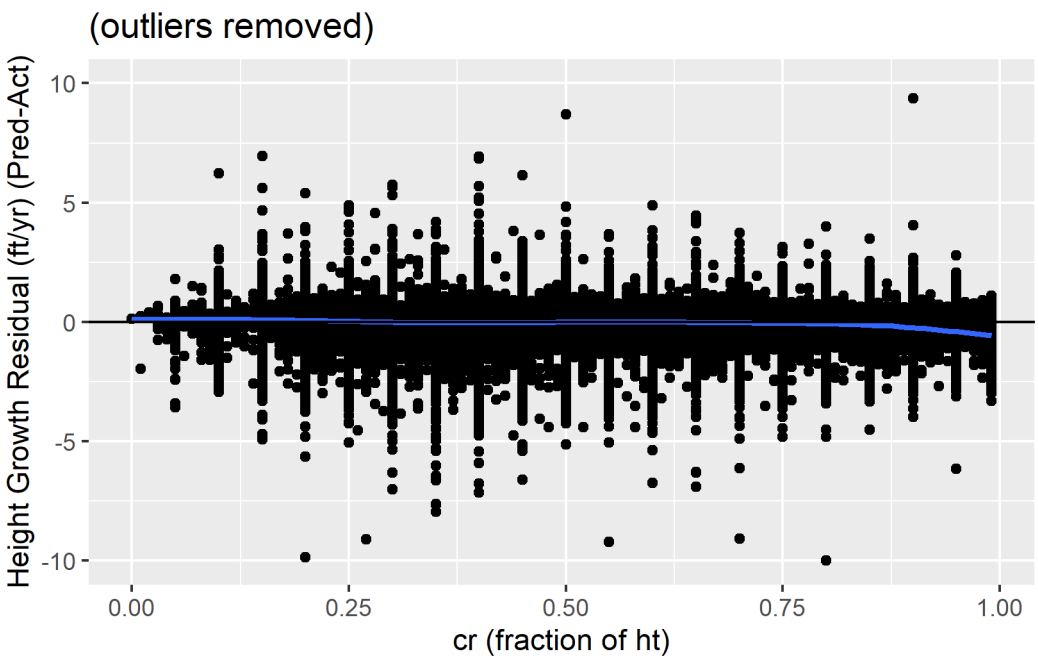
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
200
400
600
CCFL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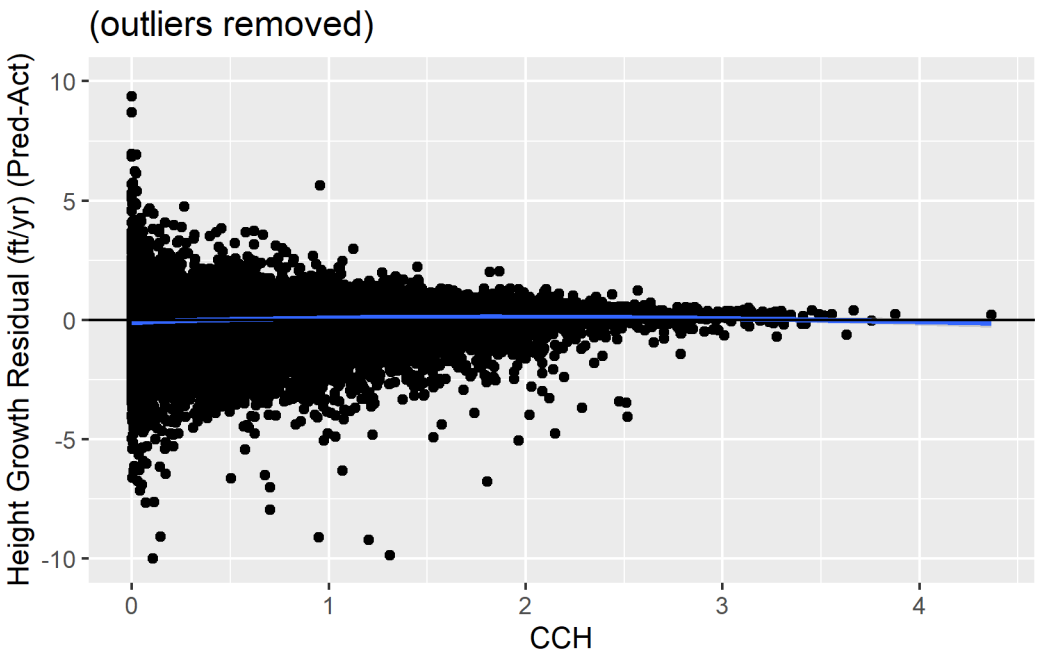
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0.00
0.25
0.50
0.75
1.00
cr (fraction of ht)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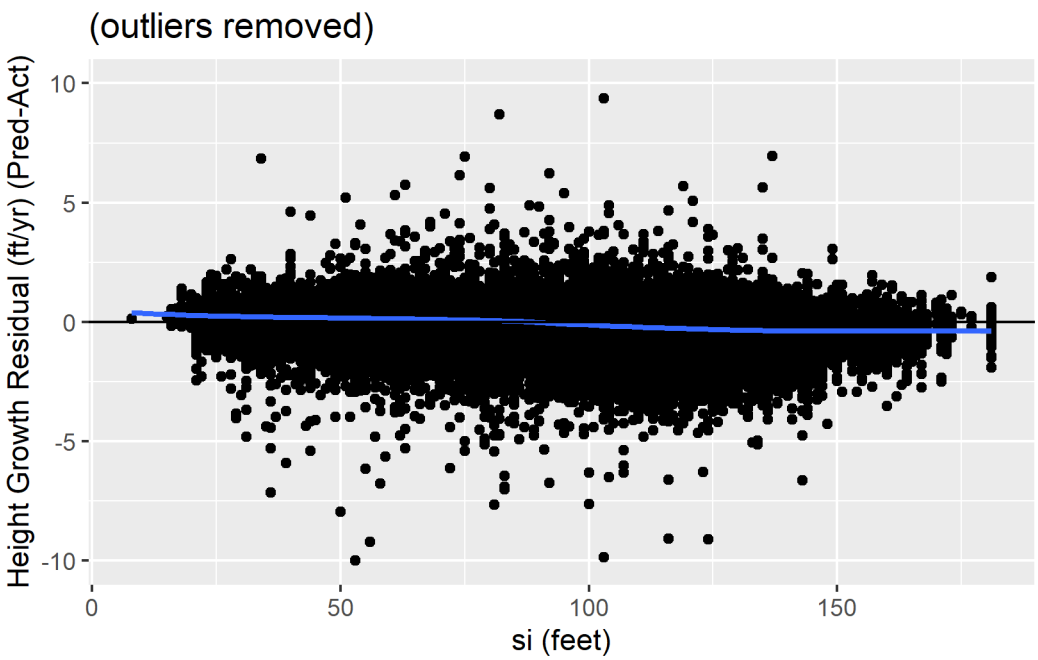
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
1
2
3
4
CCH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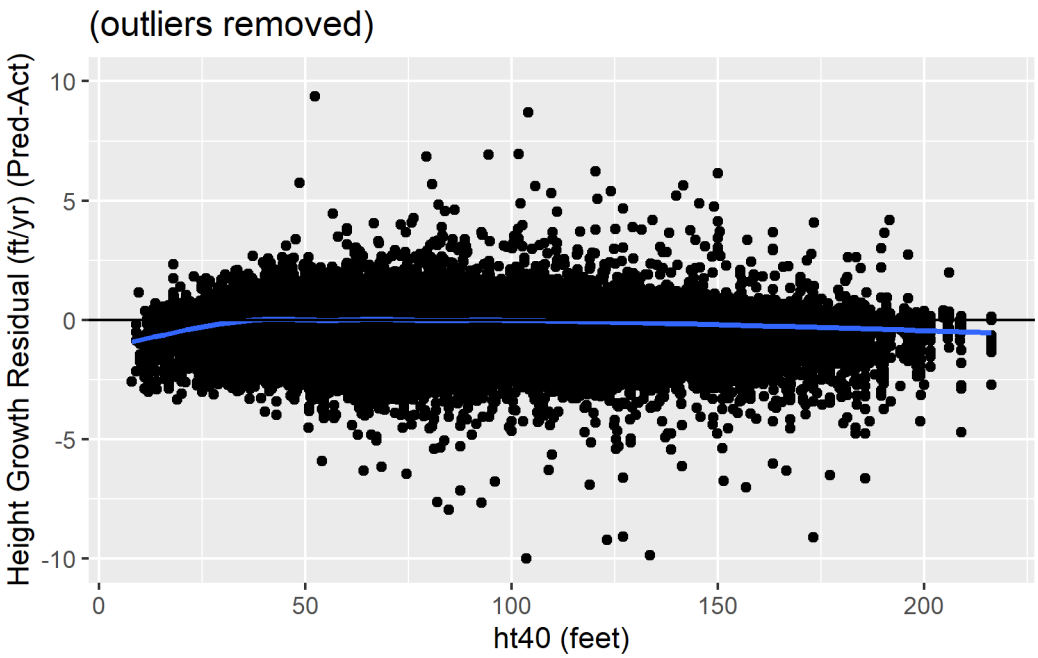
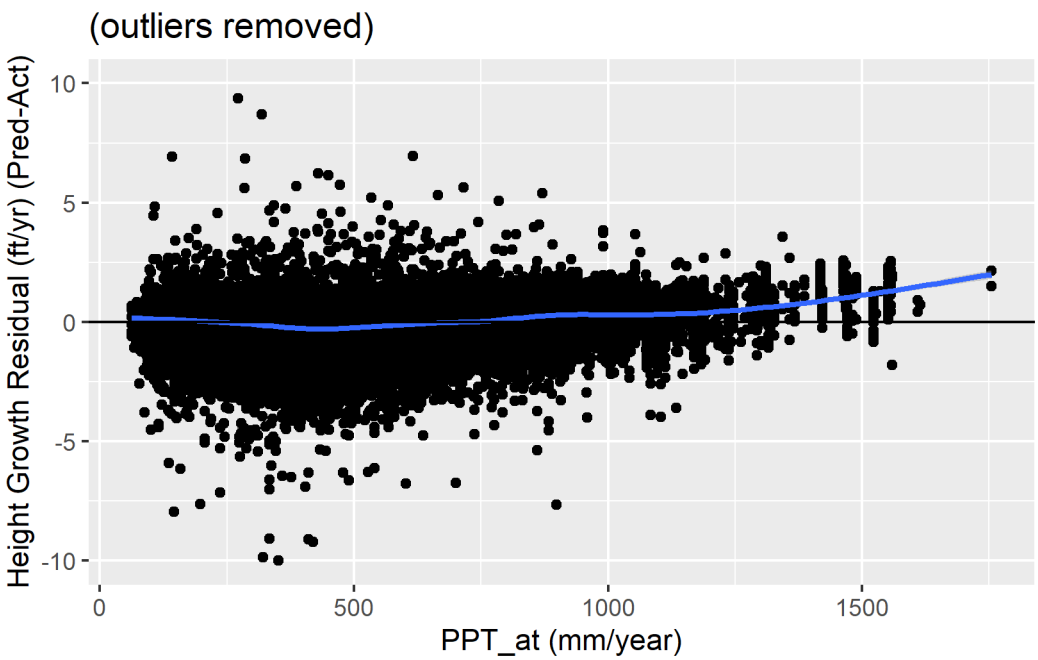
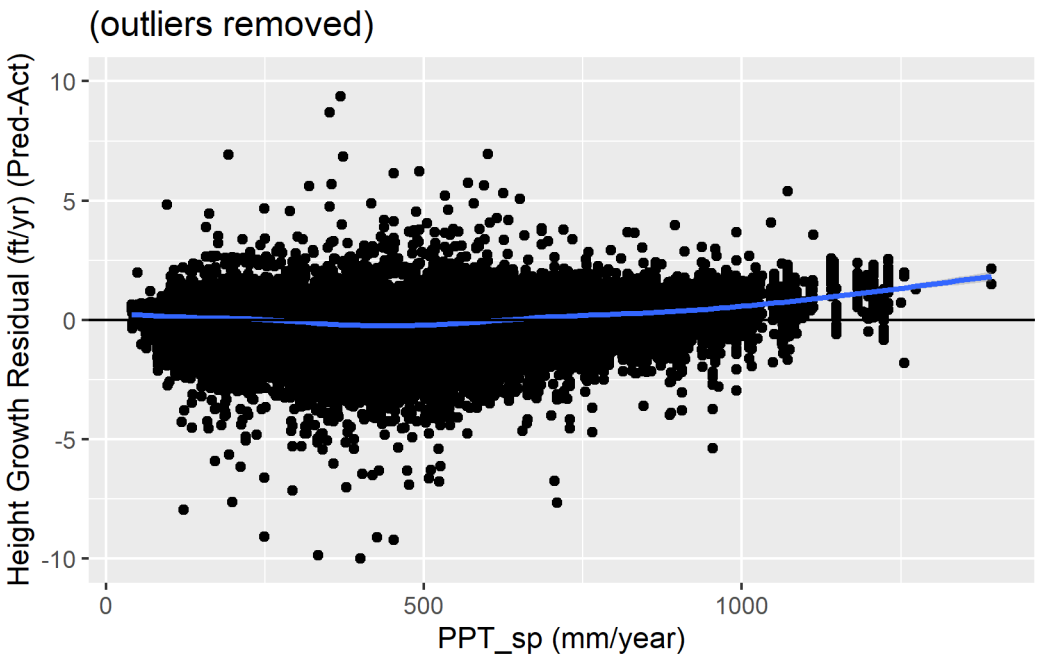
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
50
100
150
si (feet)



Environmental and Climate Variables

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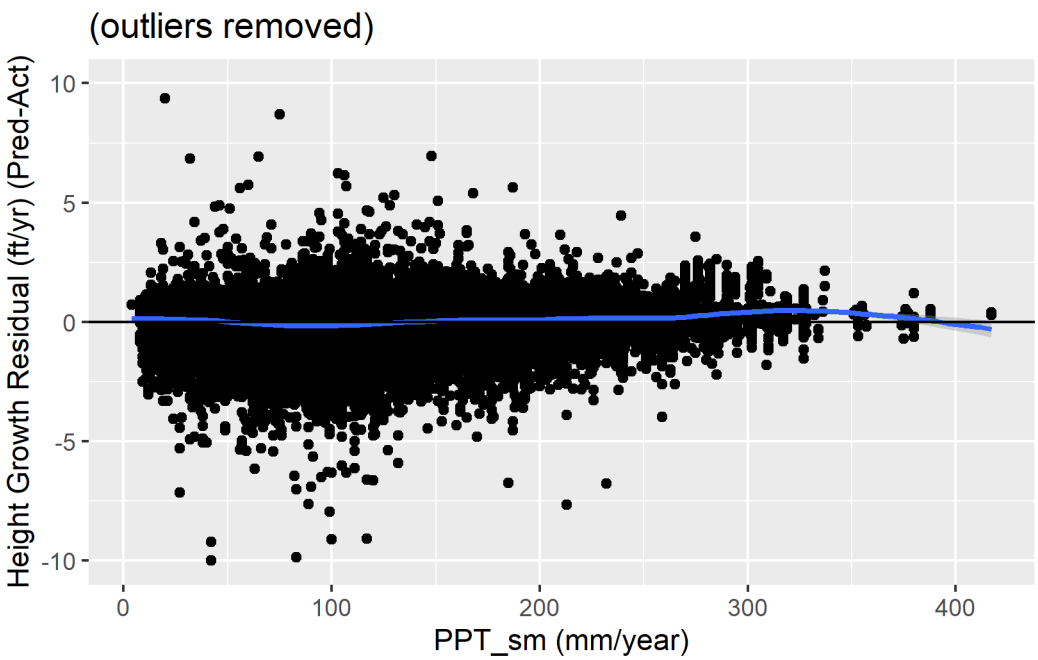
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
500
1000
PPT_sp (mm/year)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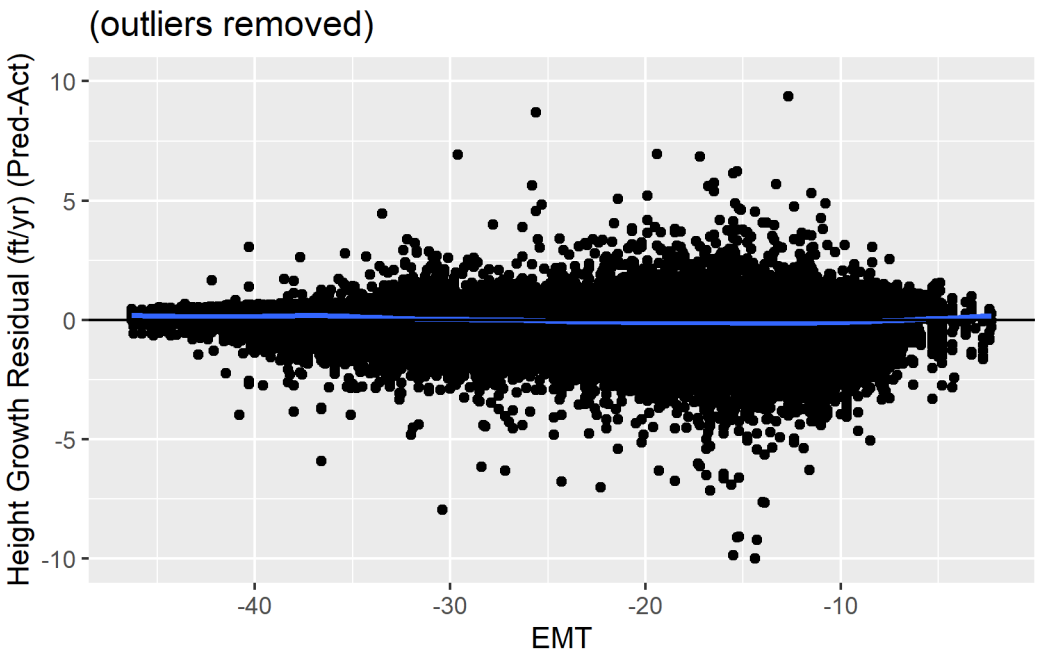
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
100
200
300
400
PPT_sm (mm/year)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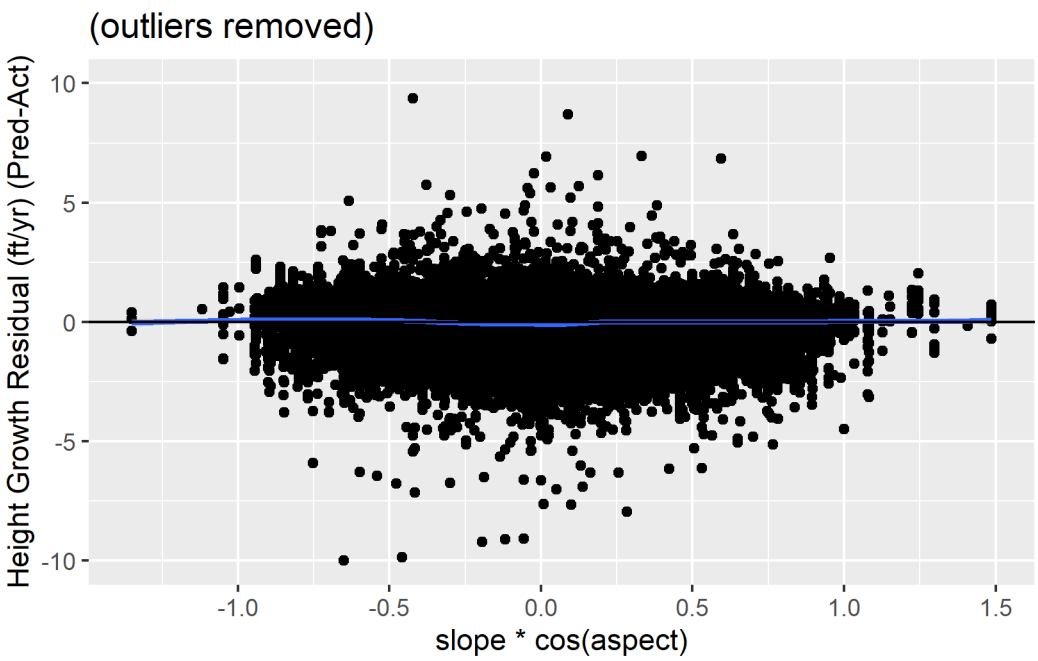
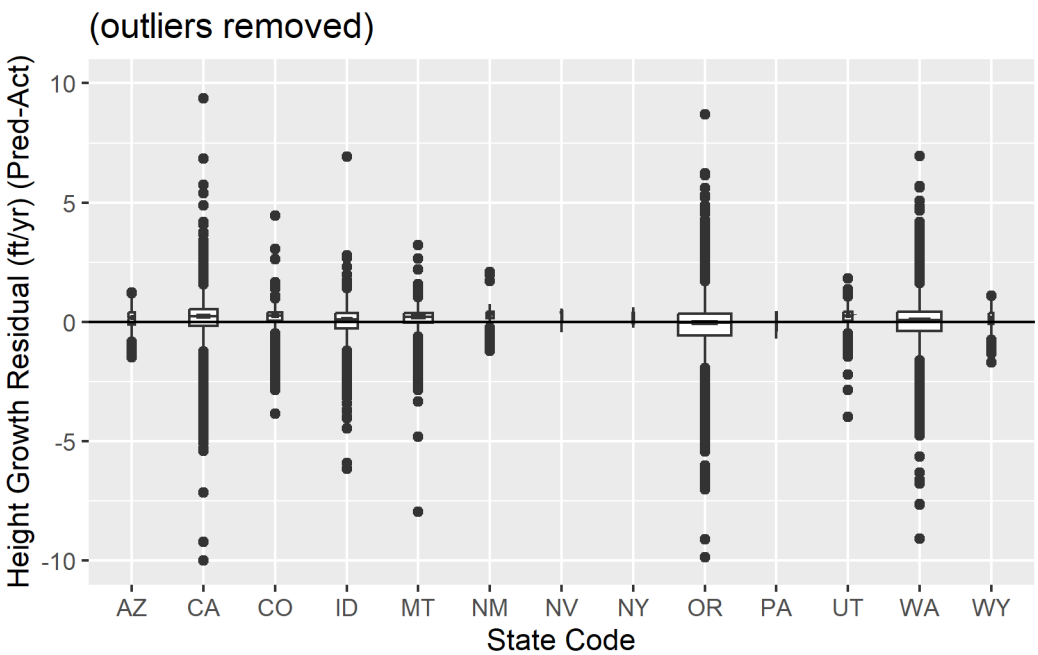
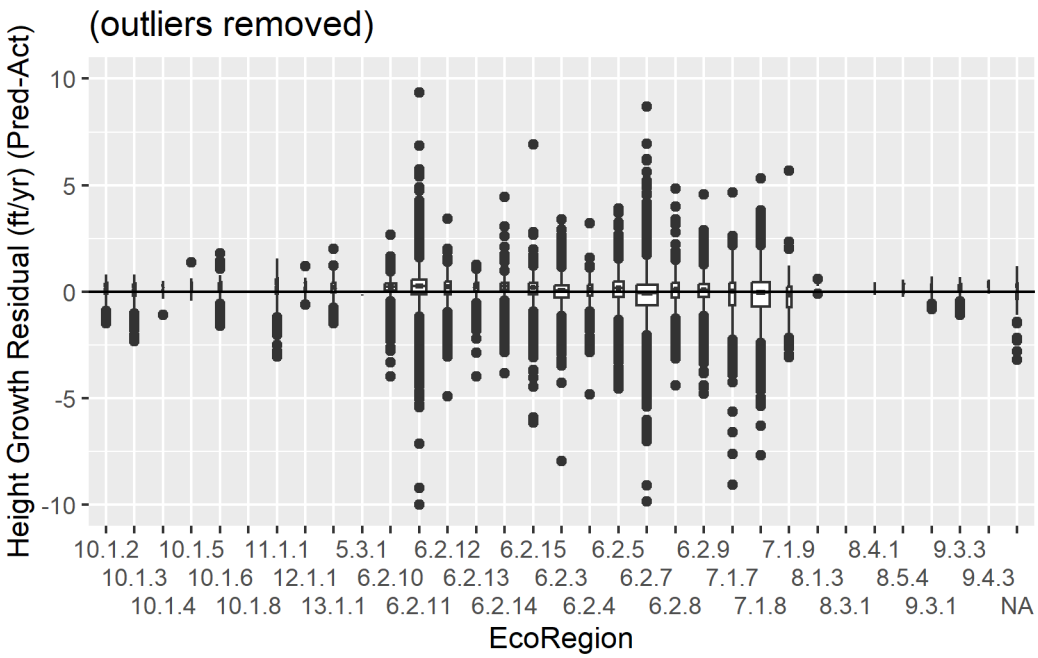
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-40
-30
-20
-10
EMT


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-1.0
-0.5
0.0
0.5
1.0
1.5
slope * cos(aspect)

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-1.0
-0.5
0.0
0.5
1.0
slope * sin(aspect)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
AZ
CA
CO
ID
MT
NM
NV
NY
OR
PA
UT
WA
WY
State Code


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
10.1.2
10.1.3
10.1.4
10.1.5
10.1.6
10.1.8
11.1.1
12.1.1
13.1.1
5.3.1
6.2.10
6.2.11
6.2.12
6.2.13
6.2.14
6.2.15
6.2.3
6.2.4
6.2.5
6.2.7
6.2.8
6.2.9
7.1.7
7.1.8
7.1.9
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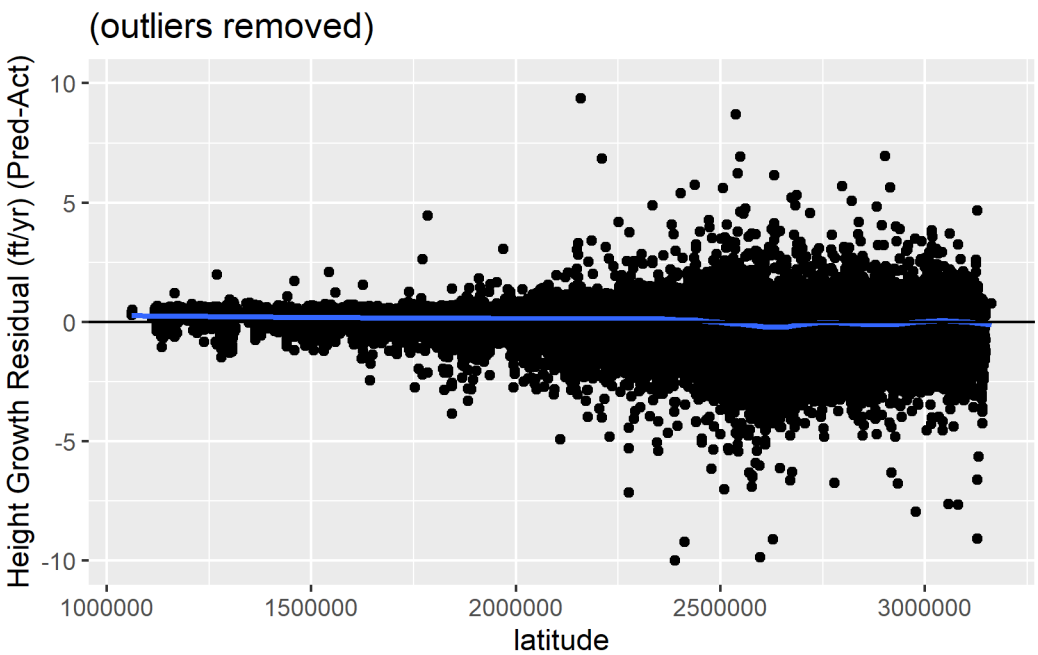
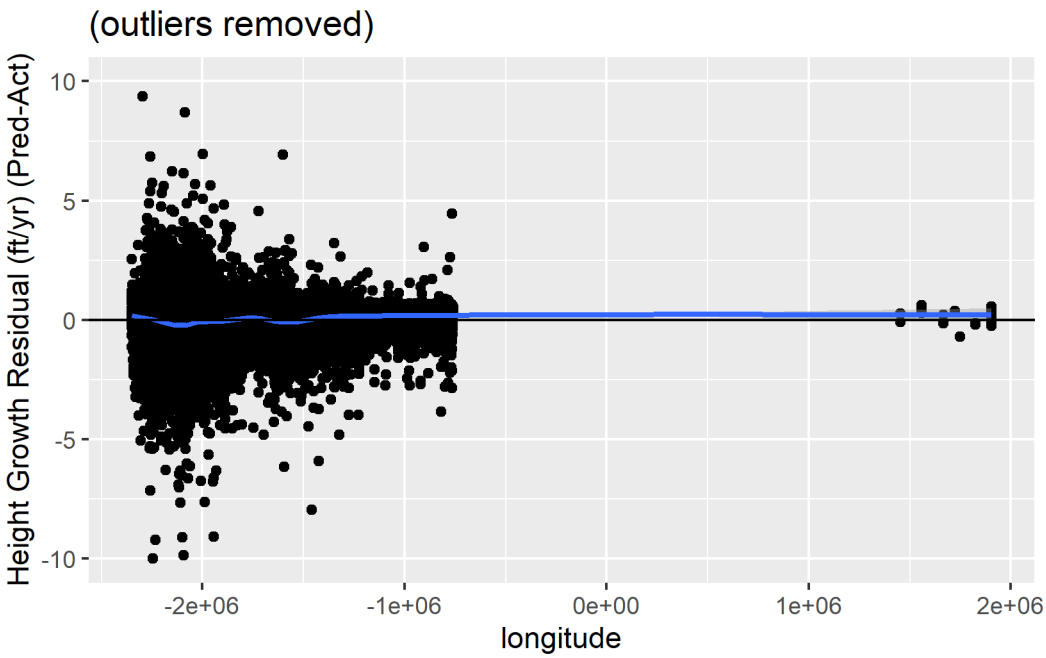
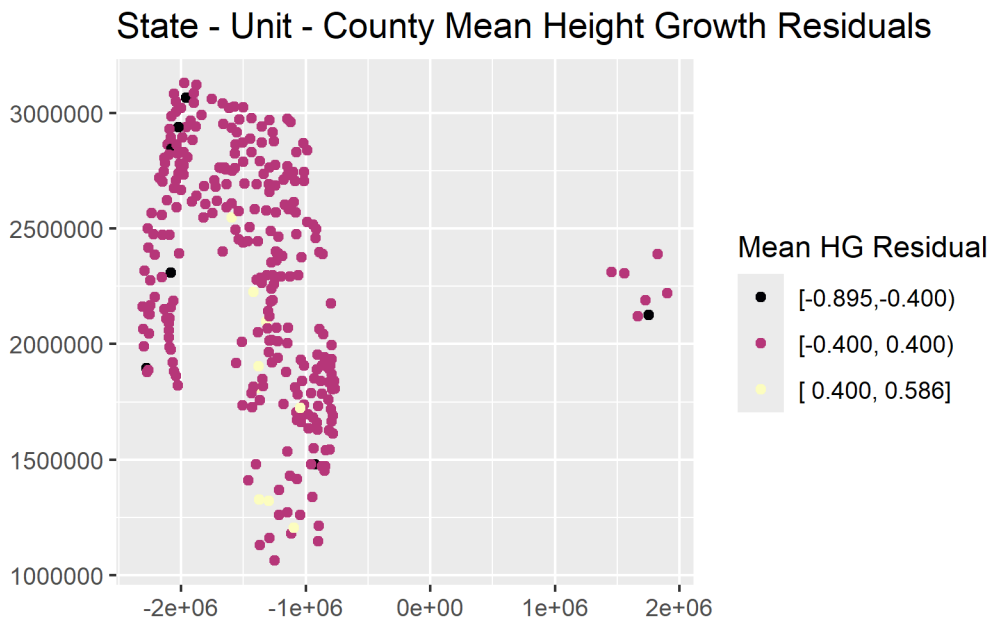
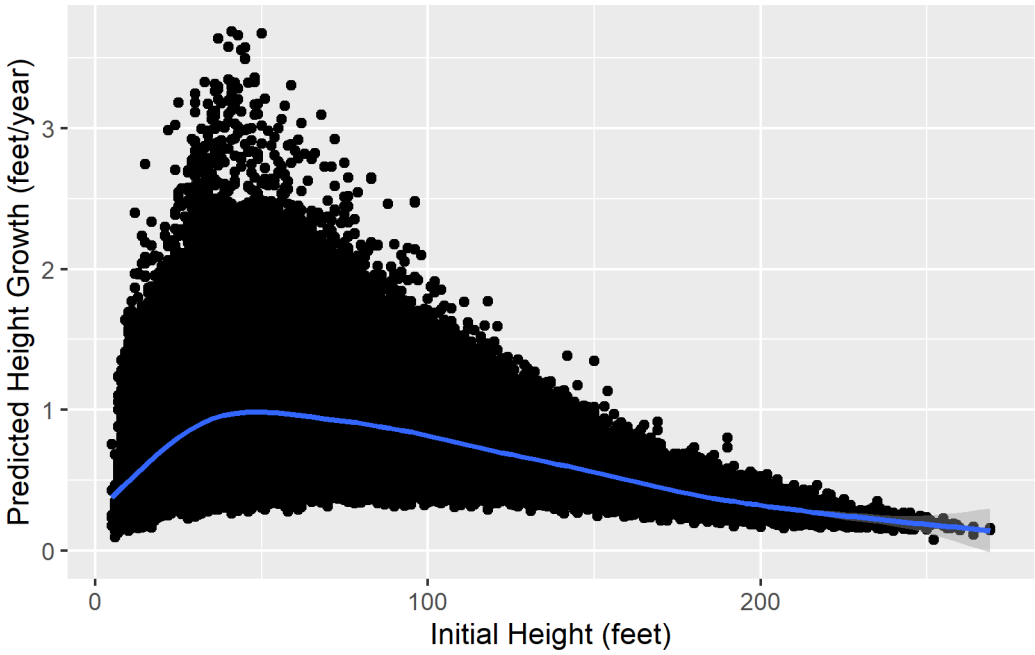
8.3.1
8.4.1
8.5.4
9.3.1
9.3.3
9.4.3
NA
EcoRegion

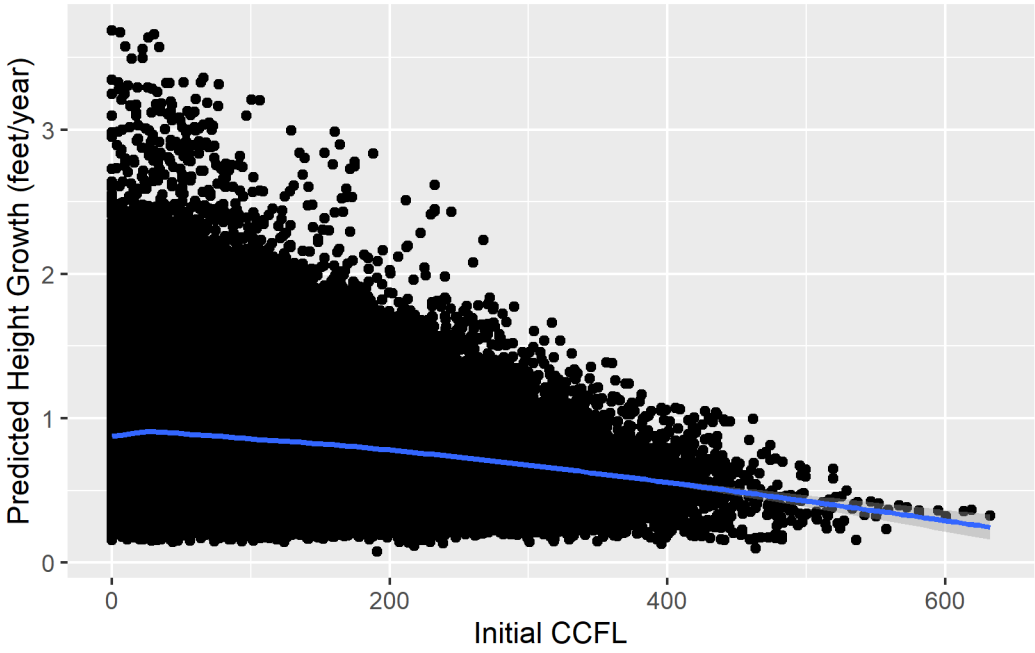

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
1000000
1500000
2000000
2500000
3000000
latitude

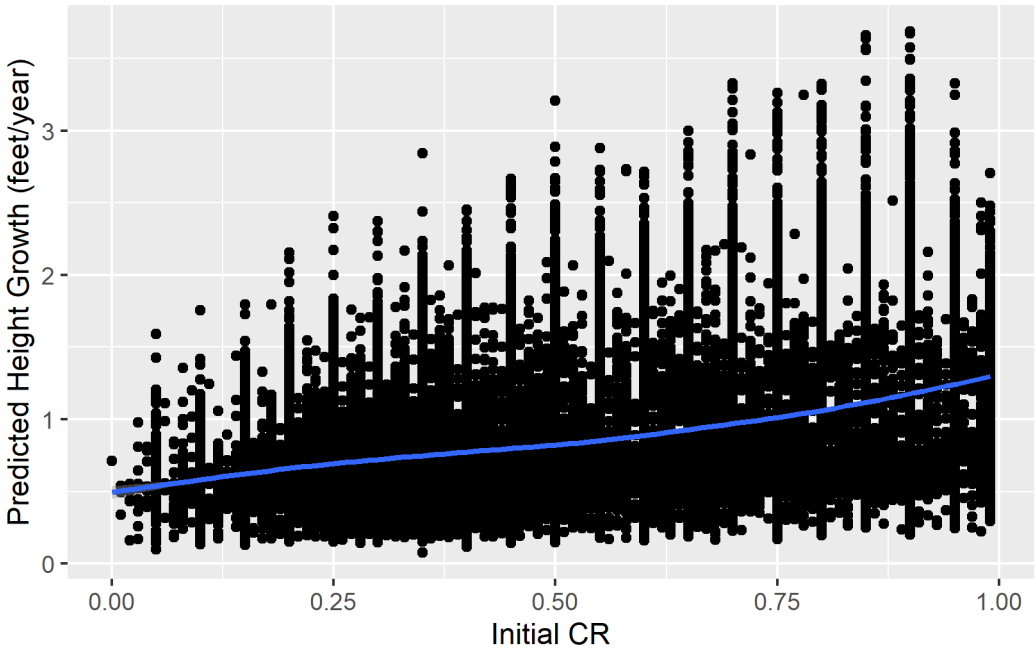

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-2e+06
-1e+06
0e+00
1e+06
2e+06
longitude

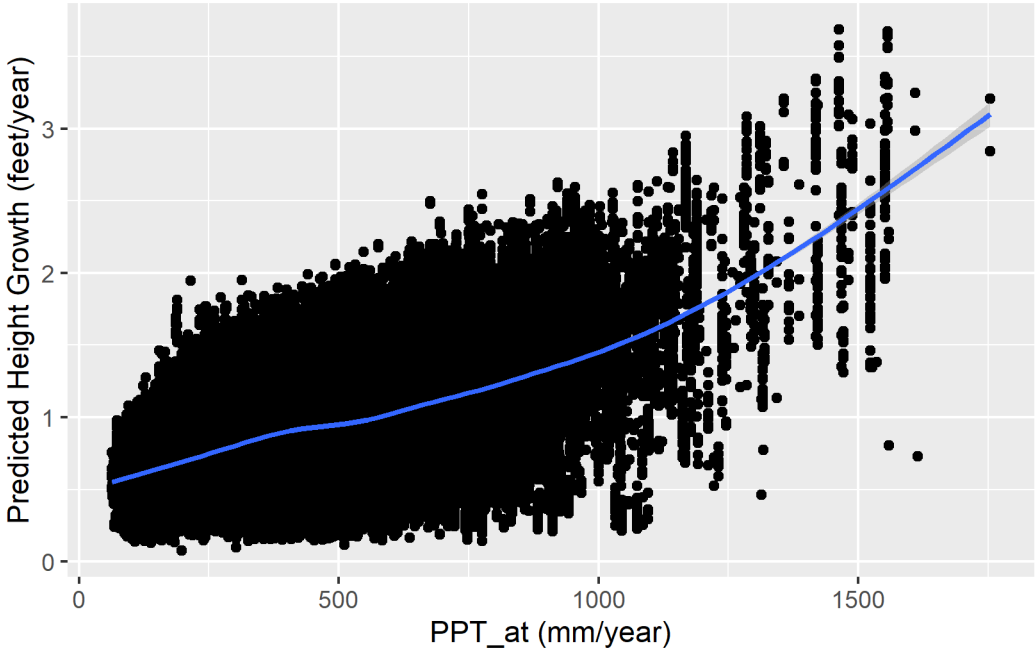

State - Unit - County Mean Height Growth Residuals
3000000
2500000
2000000
1500000
1000000
-2e+06
-1e+06
0e+00
1e+06
2e+06
Mean HG Residual
[-0.895,-0.400)
[-0.400, 0.400)
[0.400, 0.586]

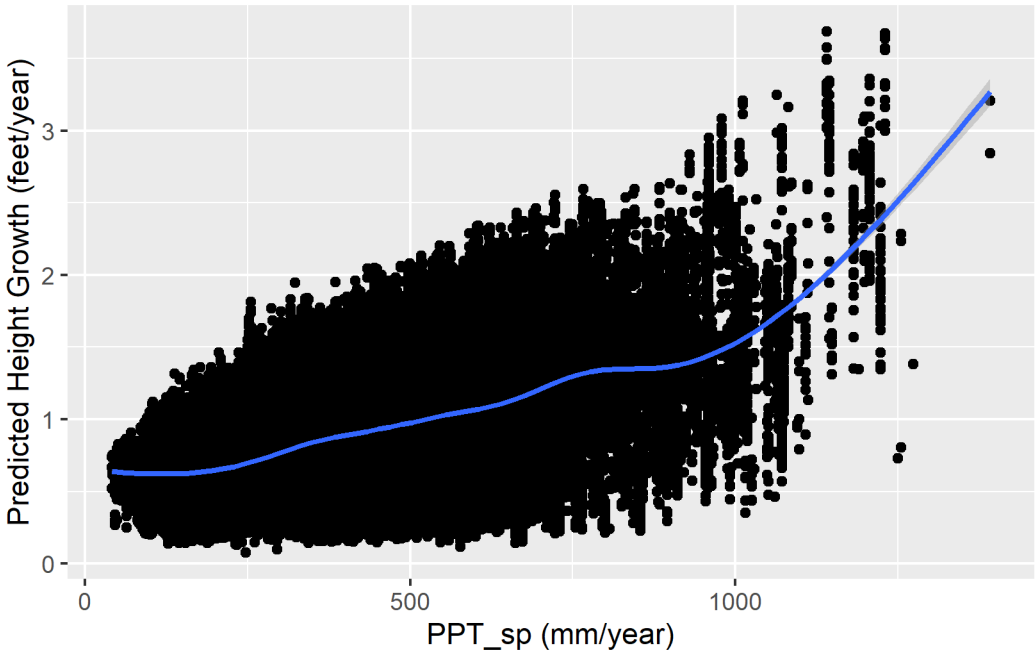
Prediction Performance

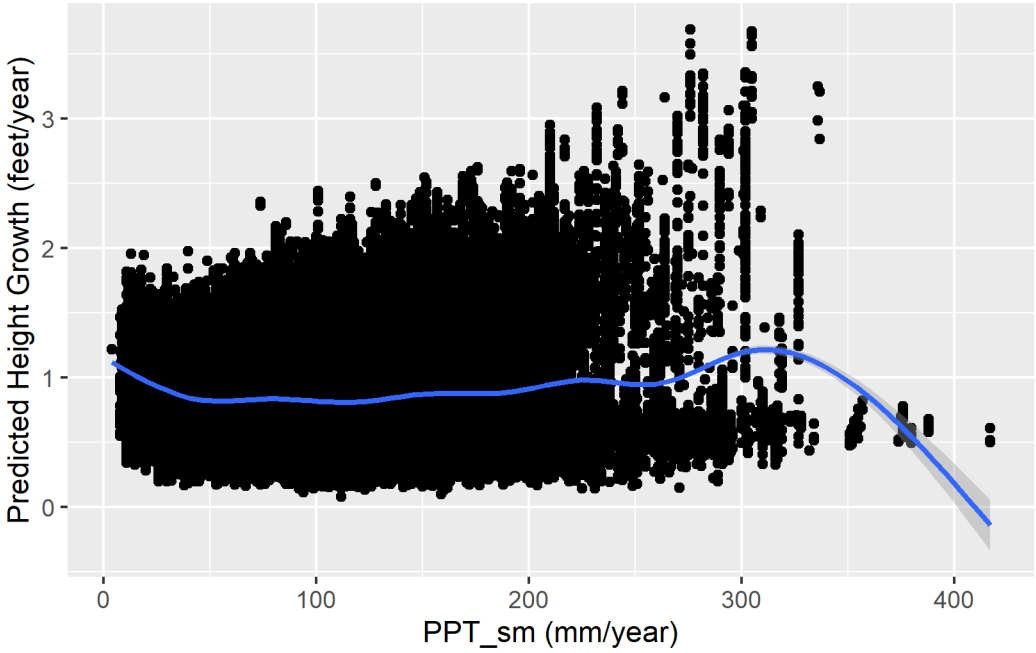


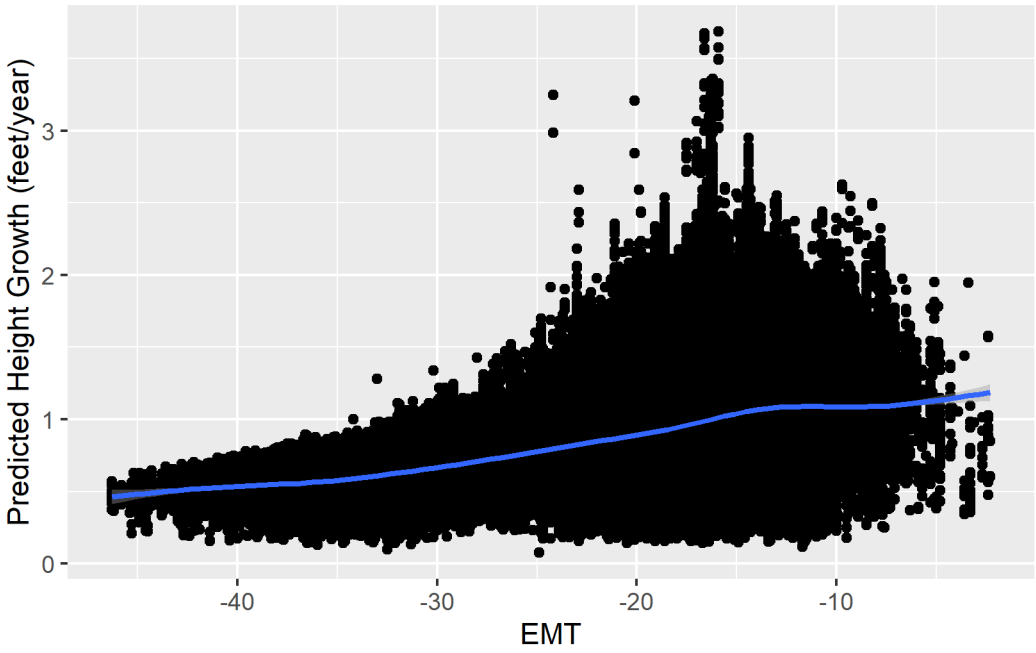



Predicted Height Growth (feet/year)
0
1
2
3
0.00
0.25
0.50
0.75
1.00
Initial CR


Predicted Height Growth (feet/year)
0
1
2
3
0
500
1000
1500
PPT_at (mm/year)


Predicted Height Growth (feet/year)
0
1
2
3
0
500
1000
PPT_sp (mm/year)


Predicted Height Growth (feet/year)
0
1
2
3
0
100
200
300
400
PPT_sm (mm/year)


Predicted Height Growth (feet/year)
0
1
2
3
-40
-30
-20
-10
EMT